

contenido.yaml

```
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#
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id: 31110130
acronym: AAIL
study_materials:
  - id: isl
    type: book
    title: An Introduction to Statistical Learning
    authors:
      - G. James
      - D. Witten
      - T. Hastie
      - R. Tibshirani
    url: https://www.statlearning.com
  - id: esl
    type: book
    title: The Elements of Statistical Learning
    edition: 2
    authors:
      - Trevor Hastie
      - Robert Tibshirani
      - Jerome Friedman
    publisher: Springer
    publication_date: 2009
    publication_place: New York
    url: https://web.stanford.edu/~hastie/ElemStatLearn/
  - id: urf
    type: article
    title: "Understanding Random Forests: from Theory to Practice"
    authors: Gilles Loupe
    url: https://arxiv.org/abs/1407.7502
    publisher: Univ. de Liège
  - type: article
    title: What Is Bootstrapping in Statistics?
    authors:
      - Courtney Taylor
    publisher: ThoughtCo
    publication_date: 2019
    url: https://www.thoughtco.com/what-is-bootstrapping-in-statistics-3126172
  - id: bfa
    type: book
    title: "Boosting: Foundations and Algorithms"
```

```

authors:
- Robert E. Schapire
- Yoav Freund
url: https://direct.mit.edu/books/oa-monograph/5342/BoostingFoundations-and-Algorithms
- id: pmlh20
  type: book
  title: Practical machine learning with H2O
  authors: Darren Cook
  publisher: O'Reilly
  url: https://www.oreilly.com/library/view/practical-machine-learning/9781491964590/
- id: clx
  title: CatBoost vs. LightGBM vs. XGBoost
  url: https://towardsdatascience.com/catboost-vs-lightgbm-vs-xgboost-c80f40662924/
- id: cct
  type: article
  title: "Classifiers Combination Techniques: A Comprehensive Review"
  authors:
    - M. Mohandes
    - M. Deriche
    - S.O. Aliyu
  url: https://ieeexplore.ieee.org/document/8335271
- id: emula
  type: article
  title: Evaluation Metrics for Unsupervised Learning Algorithms
  authors:
    - Julio-Omar Palacio-Niño
    - Fernando Berzal
  url: https://doi.org/10.48550/arXiv.1905.05667

# Práctica tema 0: Modelos múltiples
- id: 0_slides_fae
  type: slideshow
  title: Fundamentos de algoritmos Ensemble
  authors:
    - Manuel Castillo-Cara
    - Luis Sarro
  publisher: ETSII UNED
  url: https://agora.uned.es/mod/resource/view.php?id=704388
- id: 0_vid_iae
  type: video
  title: Teoría - Introducción a los algoritmos ensembles
  url: https://agora.uned.es/mod/url/view.php?id=704391
- id: 0_vid_meae
  type: video
  title: Teoría -De la Mezcla de expertos a los algoritmos ensembles

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url: https://agora.uned.es/mod/url/view.php?id=704392
- id: 0_vid_ovrovo
  type: video
  title: 1. One-vs-Rest y One-vs-One
  url:
    - https://agora.uned.es/mod/url/view.php?id=704393
    - https://agora.uned.es/mod/url/view.php?id=704394
    - https://agora.uned.es/mod/url/view.php?id=704395
- id: 0_vid_csce
  type: video
  title: 2. Códigos de salida con corrección de errores
  url:
    - https://agora.uned.es/mod/url/view.php?id=704396
    - https://agora.uned.es/mod/url/view.php?id=704397
- id: 0_vid_mrms
  type: video
  title: 3. Modelos de regresión de múltiples salidas
  url:
    - https://agora.uned.es/mod/url/view.php?id=704398
    - https://agora.uned.es/mod/url/view.php?id=704399
- id: 0_vid_scd
  type: video
  title: 4. Selección de clasificador dinámico
  url:
    - https://agora.uned.es/mod/url/view.php?id=704400
    - https://agora.uned.es/mod/url/view.php?id=704401
    - https://agora.uned.es/mod/url/view.php?id=704402
- id: 0_vid_sdc
  type: video
  title: 5. Selección dinámica de conjuntos
  url:
    - https://agora.uned.es/mod/url/view.php?id=704403
    - https://agora.uned.es/mod/url/view.php?id=704404
    - https://agora.uned.es/mod/url/view.php?id=704405

#Práctica tema 1: Bagging
- id: 1_slides_bbs
  type: slideshow
  title: Bagging, Boosting y Stacking
  authors:
    - Manuel Castillo-Cara
    - Luis Sarro
  url: https://agora.uned.es/mod/resource/view.php?id=704412
- id: 1_vid_bbs
  type: video
  title: Teoría - Bagging, Boosting y Stacking
  url: https://agora.uned.es/mod/url/view.php?id=704415

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- id: 1_vid_bagging
title: 2. Bagging
url:
 - <https://agora.uned.es/mod/url/view.php?id=704416>
 - <https://agora.uned.es/mod/url/view.php?id=704417>
 - <https://agora.uned.es/mod/url/view.php?id=704418>
- id: 1_vid_rs
title: 3. Random Subspace
url:
 - <https://agora.uned.es/mod/url/view.php?id=704419>
 - <https://agora.uned.es/mod/url/view.php?id=704420>
- id: 1_vid_fss
title: 4. Feature Selection Subspace
url:
 - <https://agora.uned.es/mod/url/view.php?id=704421>
 - <https://agora.uned.es/mod/url/view.php?id=704422>
- id: 1_vid_rf
title: 6. Random Forest
url:
 - <https://agora.uned.es/mod/url/view.php?id=704423>
 - <https://agora.uned.es/mod/url/view.php?id=704424>
- id: 1_vid_et
video: ExtraTrees
title: 7. ExtraTrees
url:
 - <https://agora.uned.es/mod/url/view.php?id=704425>
 - <https://agora.uned.es/mod/url/view.php?id=704426>
- id: 1_vid_dtb
title: 8. Data Transform Bagging
url: <https://agora.uned.es/mod/url/view.php?id=704427>
- id: 1_book_sml
type: book
title: Statistical Machine Learning
authors:
 - Yiqiao YIN
- id: 1_notes_em
type: notes
title: Ensemble methods
authors:
 - Julia Lenc
- id: 1_web_rfo
type: website
title: Random Forest y OOB
url: <https://cienciadatos.net/documentos/py36-grid-search-random-forest-out-of-bag-er>
- id: 1_web_crf
type: website

```

title: CART y Random Forest
url: https://bookdown.org/content/2031/
- id: 1_web_rfa
  type: website
  title: Random Forest Algorithm
  url: https://www.simplilearn.com/tutorials/machine-learning-tutorial/random-forest-algor
- id: 1_notes_dtcs
  type: notes
  title: Decision Tree Cheat Sheet
- id: 1_notes_bagging
  type: notes
  title: Bagging

# Práctica tema 2: Boosting
- id: 2_vid_adaboost
  type: video
  title: 1. AdaBoost
  url:
    - https://agora.uned.es/mod/url/view.php?id=704438
    - https://agora.uned.es/mod/url/view.php?id=704439
    - https://agora.uned.es/mod/url/view.php?id=704440
- id: 2_vid_gb
  type: video
  title: 2. Gradient Boosting
  url:
    - https://agora.uned.es/mod/url/view.php?id=704441
    - https://agora.uned.es/mod/url/view.php?id=704442
- id: 2_vid_xgb
  type: video
  title: 3. XGBoost
  url:
    - https://agora.uned.es/mod/url/view.php?id=704443
    - https://agora.uned.es/mod/url/view.php?id=704444
- id: 2_vid_lgbm
  type: video
  title: 4. LightGBM
  url:
    - https://agora.uned.es/mod/url/view.php?id=704445
    - https://agora.uned.es/mod/url/view.php?id=704446
- id: 2_notes_ab
  type: notes
  title: Adaptative Boosting
  authors:
    - Chaitanya Kunapareddi
- id: 2_book_cml
  type: book
  title: Combining Multiple Learners
- id: 2_inb_ab
  type: interactive_notebook

```

```

title: Algoritmos Boosting
authors: Nikita Prasad
url: https://github.com/nikitaprasad21/ML-Cheat-Codes/blob/main/Voting-Ensemble/gradient

# Práctica 3: Stacking
- id: 3_vid_voting
  type: video
  title: 1. Voting - Parte 1 URL
  authors:
    - Manuel Castillo-Cara
  url:
    - https://unedo365-my.sharepoint.com/:v:/g/personal/manuelcastillo_dia_uned_es/IQAmJCRXI
    - https://unedo365-my.sharepoint.com/:v:/g/personal/manuelcastillo_dia_uned_es/IQCcMQBa
- id: 3_vid_wa
  title: 2. Weighted Average - Parte 1 URL
  url:
    - https://agora.uned.es/mod/url/view.php?id=704457
    - https://agora.uned.es/mod/url/view.php?id=704458
    - https://agora.uned.es/mod/url/view.php?id=704459
- id: 3_vid_smc
  title: 3. Seleccion Miembros Conjunto
  url:
    - https://agora.uned.es/mod/url/view.php?id=704460
    - https://agora.uned.es/mod/url/view.php?id=704461
    - https://agora.uned.es/mod/url/view.php?id=704462
- id: 3_vid_stacking
  title: 4. Stacking
  url:
    - https://agora.uned.es/mod/url/view.php?id=704463
    - https://agora.uned.es/mod/url/view.php?id=704464
- id: 3_vid_blending
  title: 5. Blending
  url:
    - https://agora.uned.es/mod/url/view.php?id=704465
    - https://agora.uned.es/mod/url/view.php?id=704466
- id: 3_vid_sl
  title: 6. Super Learner - Teoría
  url: https://agora.uned.es/mod/url/view.php?id=704467
- id: 3_vid_slp
  type: video
  title: 6.1. Super Learner - Práctica
- id: 3_vid_sldr
  type: video
  title: 6.2. Super Learner - Dataset real
  url: https://agora.uned.es/mod/url/view.php?id=704469
- id: 3_web_scss

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type: website
title: "StackingClassifier: Simple stacking"
url: https://rasbt.github.io/mlxtend/user_guide/classifier/StackingClassifier/
- id: 3_vid_ae
  title: Algoritmos ensamblados
  url: https://www.cs.us.es/~fsancho/?e=106
- id: sl
  type: article
  title: Super Learner
  authors:
    - Eric C. Polley
    - Mark J. van der Laan
    - Alan E. Hubbard
- id: slp
  type: article
  title: Super Lerner in Prediction
  authors:
    - Eric C. Polley
    - Mark J. van der Laan

# Práctica 4: Aprendizaje no supervisado
- id: 4_slides_ans
  type: slideshow
  title: 8. Aprendizaje No Supervisado [Presentación]
- id: 4_notes_ans
  type: notes
  title: 8. Aprendizaje No Supervisado [Apuntes]
- id: 4_slides_pca
  type: slideshow
  title: 8.2. PCA
- id: 4_vid_ans
  type: video
  title: 8.1. Aprendizaje No Supervisado - Teoría
  url: https://agora.uned.es/mod/url/view.php?id=704483
- id: 4_vid_ac
  type: video
  title: 8.2. Algoritmos clustering
  url:
    - https://agora.uned.es/mod/url/view.php?id=704484
    - https://agora.uned.es/mod/url/view.php?id=704485
- id: 4_vid_iac
  title: 8.4. Implementacion Algoritmos clustering
  url:
    - https://agora.uned.es/mod/url/view.php?id=704486
    - https://agora.uned.es/mod/url/view.php?id=704487
- id: 4_vid_km

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    title: 8.6. k-Means
    url: https://agora.uned.es/mod/url/view.php?id=704488
- id: 4_vid_cj
  title: 8.7. Clustering Jerarquico
  url: https://agora.uned.es/mod/url/view.php?id=704489
- id: 4_vid_mbd
  title: 8.8. Metodos basado en densidad
  url: https://agora.uned.es/mod/url/view.php?id=704490
- id: 4_vid_dc
  title: 8.9. Determinar clusters
  url: https://agora.uned.es/mod/url/view.php?id=704491
- id: 4_vid_cd
  title: 8.10. Conjunto de datos
  url: https://agora.uned.es/mod/url/view.php?id=704492
- id: 4_vid_tdc
  title: 8.11. Tecnicas para determinar clusters
  url: https://agora.uned.es/mod/url/view.php?id=704493
- id: 4_vid_agglomerative
  title: 8.12. Agglomerative
  url: https://agora.uned.es/mod/url/view.php?id=704494
- id: 4_vid_ikm
  title: 8.13. Implementación de k-Means URL
  url: https://agora.uned.es/mod/url/view.php?id=704495
- id: 4_vid_ms
  title: 8.14. Mean Shift
  url: https://agora.uned.es/mod/url/view.php?id=704496
- id: 4_vid_ra
  title: 8.15. Reglas de asociacion
  url: https://agora.uned.es/mod/url/view.php?id=704497
- id: 4_vid_aa
  type: video
  title: 8.16. Algoritmo Apriori
  url: https://agora.uned.es/mod/url/view.php?id=704498
- type: website
  title: Página de Pier Luca Lanzi
  url: http://www.pierlucalanzi.net/
- title: Explicación de ULAs
  url: https://ligdigonzalez.com/todo-sobre-aprendizaje-no-supervisado-en-machine-learning/
- title: ULAs Vs. SLAs
  url: https://www.cs.us.es/~fsancho/?e=77
- title: Detección de Anomalías URL
  url: https://elbaultdelprogramador.com/aprendizaje-nosupervisado-clustering/
- title: "[Código] Algoritmos URL"
  url: https://builtin.com/data-science/unsupervised-learning-python
- title: Tipos de aprendizaje no supervisado URL
  url: https://medium.com/datos-y-ciencia/aprendizaje-no-supervisado-en-machine-learning-a

```

- type: cheatsheet
 - title: "VIP Cheatsheet: Unsupervised Learning"
 - authors:
 - Afshine Amidi
 - Shervine Amidi
 - url: <https://stanford.edu/~shervine>
- type: notes
 - title: Understanding DBSCAN Algorithm
 - publisher: Analytics Vidhya
- type: slideshow
 - title: "Hierarchical Clustering: Relationship between Clusters"
- type: image
 - title: Algoritmos de Clustering

units:

- id: 0
 - title: Modelos múltiples
 - sections:
 - title: Introducción a los algoritmos ensembles
 - study_materials:
 - id_ref: 0_slides_fae
 - id_ref: 0_vid_iae
 - title: De la Mezcla de expertos a los algoritmos ensembles
 - study_materials:
 - id_ref: 0_vid_meae
 - title: 1. One-vs-Rest y One-vs-One
 - study_materials:
 - id_ref: 0_vid_ovrovo
 - title: 2. Códigos de salida con corrección de errores
 - study_materials:
 - id_ref: 0_vid_csce
 - title: 3. Modelos de regresión de múltiples salidas
 - study_materials:
 - id_ref: 0_vid_mrms
 - title: 4. Selección de clasificador dinámico
 - study_materials:
 - id_ref: 0_vid_scd
 - title: 5. Selección dinámica de conjuntos
 - study_materials:
 - id_ref: 0_vid_sdc
 - title: Bagging, Boosting y Stacking
 - study_materials:
 - id_ref: 1_slides_bbs
 - id_ref: 1_vid_bbs
- id: 1

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title: Bosques aleatorios
sections:
- id: 1.1
  title: Descomposición en sesgo/varianza
  sections:
  - title: Error predicción esperado
  tags:
  - error-prediccion-esperado
  study_materials:
  - id_ref: isl
    reference: 2.2.2
  - id_ref: esl
    reference: 7.1-7.3
    pags: 219-228
  - id_ref: urf
    reference: Section 4.1 "Random forests"
    pages: 55-60
- id: 1.2
  title: El método bootstrap
  tags:
  - bootstrap
  - bootstrapping
  study_materials:
  - id_ref: wib
  - id_ref: isl
    reference: Section 5.2
- id: 1.3
  title: "Agregación de Bootstrap: Bagging"
  tags:
  - bootstrap-aggregating
  - bagging
  study_materials:
  - id_ref: isl
    reference: Section 8.2.1
  - id_ref: esl
    reference: Section 8.7
    pages: 282-288
  - id_ref: esl
    reference: Section 8.7.1
  - id_ref: 1_slides_bbs
    reference: Apartados Método Bootstrap y Bootstrap Aggregation
  - id_ref: 1_vid_bagging
  sections:
  - title: Error OOB
    tags:
    - error-oob

```

- title: Random Subspace
 - study_materials:
 - id_ref: 1_vid_rs
 - tags:
 - random-subspace
- title: Feature Selection Subspace
 - study_materials:
 - id_ref: 1_vid_fss
 - tags:
 - feature-selection-subspace
- id: 1.4
 - title: Bosques aleatorios
 - tags:
 - bosque-aleatorio
 - random-forest
 - study_materials:
 - id_ref: 1_vid_rf
- title: ExtraTrees
 - tags:
 - extratree
 - study_materials:
 - id_ref: 1_vid_et
- title: Data Transform Bagging
 - study_materials:
 - id_ref: 1_vid_dtb
 - tags:
 - data-transform-bagging
- id: 2
 - title: Intensificación (anteriormente, potenciación ("Boosting") y árboles aditivos)
 - sections:
 - id: 2.1
 - title: Introducción
 - study_materials:
 - id_ref: isl
 - reference: Apartado 8.2.3
 - id_ref: 1_slides_bbs
 - reference: Apartado Boosting
 - id: 2.2
 - title: AdaBoost
 - tags:
 - adaboost
 - study_materials:
 - id_ref: bfa
 - reference: 1.2
 - pages: 4-7
 - id_ref: bfa

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    reference: 1.2.1
    pages: 7-11
  - id_ref: BFA
    reference: 1.2.3
    pages: 11-14
  - id_ref: 2_vid_adaboost
- id: 2.3
  title: Intensificación de gradiente
  tags: intensificacion-gradiente
  study_materials:
  - id_ref: esl
    reference: 10.9 Boosting trees
    pages: 353-358
  - id_ref: esl
    reference: 10.10 Numerical Optimization via Gradient Boosting
    pages: 358-361
  - id_ref: esl
    reference: 10.13 Interpretation
    pages: 367-370
  - id_ref: esl
    reference: 10.14 Illustrations
    pages: 370-380
  - id_ref: 2_vid_gb
- id: 2.4
  title: Implementaciones de intensificación de gradientes
  sections:
  - title: XGBoost
    tags:
    - xgboost
  study_materials:
  - id_ref: 2_vid_xgb
  - title: GBM
  - title:
  - title: LightGBM
    tags:
    - lightgbm
    id_ref: 2_vid_lgbm
  - title: CATBoost
- id: 2.5
  title: Prácticas de programación de intensificación
  study_materials:
  - id: pmlh20
    details: Capítulo 6 "Gradient Boosting Machines"
- id: 3
  title: Otras combinaciones de modelos (anteriormente, aprendizaje conjunto)
  sections:

```

- id: 3.1
 - title: Introducción
 - study_materials:
 - id_ref: cct
 - reference: Introducción
- id: 3.2
 - title: Marco general para la combinación de clasificadores
 - sections:
 - title: Combinación clasificadores
 - tags:
 - combinacion-clasificadores
 - study_materials:
 - id_ref: cct
 - reference: Sección II
- id: 3.3
 - title: Estrategias para la combinación de clasificadores
 - study_materials:
 - id_ref: cct
 - reference: Sección III
 - mandatory: false
 - id_ref: ccs
 - mandatory: false
 - sections:
 - title: Votación
 - tags:
 - votacion
 - voting
 - study_materials:
 - id_ref: 3_vid_voting
 - title: Weighted average
 - study_materials:
 - id_ref: 3_vid_wa
 - title: Selección Miembros Conjunto
 - tags:
 - seleccion
 - study_materials:
 - id_ref: 3_vid_smc
 - title: Apilamiento
 - tags:
 - apilamiento
 - stacking
 - study_materials:
 - id_ref: 1_slides_bbs
 - reference: Apartado Stacking
 - mandatory: true
 - id_ref: 3_vid_stacking

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        mandatory: false
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        study_materials:
          - id: 3_vid_blending
      - title: Super Learner
        study_materials:
          - id: 3_vid_sl
          - id: 3_vid_slp
          - id: 3_vid_sldr
- id: 4
  title: Aprendizaje no supervisado
  sections:
    - id: 4.1
      title: "Introducción al aprendizaje no supervisado"
      study_materials:
        - id_ref: esl
          reference: 14.1
          pages: 485-487
        - id_ref: 4_slides_ans
        - id_ref: 4_notes_ans
        - id_ref: 4_vid_ans
      sections:
        - title: Análisis de componentes principales (PCA)
          tags:
            - analisis-componentes-principales
            - pca
          study_materials:
            - id_ref: 4_slides_pca
    - id: 4.2
      title: Análisis de grupos
      tags:
        - agrupamiento
        - clustering
      study_materials:
        - id_ref: esl
          reference: 14.3
          pages: 501-527
        - id_ref: 4_vid_ac
        - id_ref: 4_vid_iac
      sections:
        - title: Clasificación no supervisada
          tags:
            - clasificacion-no-supervisada
    - id: 4.3

```

```

title: Medidas de similitud y disimilitud
sections:
- title: Matrices de proximidad
  tags:
  - matrices-proximidad
- title: Disimilitud
  tags:
  - disimilitud
  - disimilitud-confenetica
  - agrupamiento-datos-ausentes
- id: 4.4
  title: Familias de métodos de agrupamiento
- id: 4.5
  title: Métodos de clasificación no supervisada
  sections:
  - title: Agrupamiento combinatorio
    tags:
    - agrupamiento-combinatorio
  - title: K-medias
    tags:
    - k-medias
    - k-means
    study_materials:
    - id_ref: 4_vid_km
    - id_ref: 4_vid_ikm
  - title: Mezclas de Gaussianas
    tags:
    - agrupamiento-mezclas-gaussianas
    - mezcla-gaussianas
  - title: Cuantización de vectores
  - title: K-mediods
  - tags:
    - k-mediods
  - title: Agrupamiento jerárquico
    study_materials:
    - id_ref: 4_vid_cj
    sections:
    - title: Agrupamiento jerárquico aglomerativo
      tags:
      - agrupamiento-aglomerativo
      - agrupamiento-jerarquico-aglomerativo
      - clustering-jerarquico-aglomerativo
      study_materials:
      - id: 4_vid_agglomerative
    - title: Agrupamiento jerárquico divisivo
      tags:

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- agrupamiento-divisivo
- agrupamiento-jerarquico-divisivo
- clustering-jerarquico-divisivo
- sections:
 - title: Encadenamiento
 - tags:
 - encadenamiento
 - chaining
- title: Agrupamiento basados en densidad
 - tags:
 - agrupamiento-basado-densidad
 - study_materials:
 - id_ref: 4_vid_mbd
 - sections:
 - title: DBSCAN
 - tags:
 - dbscan
 - title: Mean Shift
 - tags:
 - mean-shift
 - study_materials:
 - id_ref: 4_vid_ms
- id: 4.6
 - title: Evaluación de agrupamientos
 - study_materials:
 - id_ref: emul
 - sections:
 - title: Determinar clusters
 - tags:
 - estimacion-numero-grupos
 - study_materials:
 - id_ref: 4_vid_dc
 - id_ref: 4_vid_cd
 - id_ref: 4_vid_tdc
 - title: Índice Hartigan
 - tags:
 - indice-hartigan
 - title: Índice Dunn
 - tags:
 - indice-dunn
 - title: Coeficiente silhouette
 - tags:
 - coeficiente-silhouette
- title: 8.15. Reglas de asociacion
 - study_materials:
 - id_ref: 4_vid_ra

```
sections:
- title: Algoritmo a priori
  study_materials:
    - id_ref: 4_vid_aa
- id: 4.7
  title: Reglas de asociación
  sections:
  - title: Introducción a las reglas de asociación
    tags:
      - reglas-asociacion
  - title: Algoritmo apriori
    tags:
      - algoritmo-apriori
```